16S LiFePO4 BMS - External Procurement Sourcing (Non-ETStore)

Specialized Components Procured Outside ET Store (Indian Vendors: LionCircuits, Robu.in, Evelta, Sharvi Electronics, Rajiv Electronics)

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Sourcing Scope: Non-ETStore Vendors

LionCircuits	Robu.in	Evelta	Sharvi	Rajiv	Total Non-ETStore
lioncircuits.com	robu.in	evelta.com	sharvielectronics.com	rajivelectronics.com	5 Indian Vendors
1 Item | 1 pc	5 Items | 6 pcs	7 Items | 7 pcs	1 Item | 1 pc	7 Items | 7 pcs (14 pk)	21 Items | 22 pcs
₹ 665.29	₹ 775.00	₹ 814.64	₹ 184.88	₹ 304.10 / ₹ 448.10	₹ 2,743.91*

#	Component & Ref	Component Description	Component Footprint	Qty	Cost Online	Total Cost	Website
Vendor: Lion Circuits (lioncircuits.com) - Primary Battery Front-End Monitoring IC							
1	**BQ76952PFBR** U1	16S LiFePO4 Battery Monitor, Autonomous Protector and Balancer IC (High-Accuracy Cell ADC)	Package_QFP:TQFP-48_7x7mm_P0.5mm	1	₹ 665.29	₹ 665.29	**lioncircuits.com**
Vendor: Robu.in (robu.in) - Current Sensing, High-Endurance Memory, Drivers and Gate Supply							
2	**FL-2B 100A 75mV** RS1	XIQI Electric CG FL-2B 100A 75mV Class 0.5 Precision DC Current Shunt (Chassis Bolt Mount)	Chassis_BoltMount (Kelvin: J31 / J32)	1	₹ 352.00	₹ 352.00	**robu.in**
3	**FM24CL64B-GTR** U21	Infineon 64 Kbit (8Kx8) High-Endurance Non-Volatile I2C FRAM (Replaces EEPROM for 10^14 Cycles)	Package_SO:SOIC-8_3.9x4.9mm_P1.27mm	1	₹ 232.00	₹ 232.00	**robu.in**
4	**UCC27517DBVR** U2, U3	TI 4A High-Speed Low-Side Gate Drivers (Charge and Discharge Power MOSFET Switching Banks)	Package_TO_SOT_SMD:SOT-23-5	2	₹ 35.00	₹ 70.00	**robu.in**
5	**B1209S-1WR3** U14	1W 12V-to-9V Isolated DC-DC Converter Module for Gate Driver Floating Bias Rail (SIP-4 THT)	Converter_DCDC:Converter_DCDC_Murata_MEE1SxxxxSC_THT	1	₹ 109.00	₹ 109.00	**robu.in**
6	**PESD2CAN,215** D5	Nexperia Dual-Line CAN Bus Transient Surge ESD Protection TVS Diode (CANH / CANL lines)	Package_TO_SOT_SMD:SOT-23	1	₹ 12.00	₹ 12.00	**robu.in**
Vendor: Evelta (evelta.com) - Isolated CAN Transceiver, High-Voltage Buck and Precision Protection							
7	**ISO1050DUBR** U13	TI 5V Galvanically Isolated CAN Transceiver (5000 Vrms Isolation, up to 1 Mbps, SOP-8)	Package_SO:SOP-8_6.62x9.15mm_P2.54mm	1	₹ 230.10	₹ 230.10	**evelta.com**
8	**LM5164DDA** U10	TI LM5164 100V 1A Synchronous Step-Down Buck Converter (HSOP-8-EP / 7Semi Module)	Package_SO:HSOP-8-1EP_3.9x4.9mm_P1.27mm_EP2.41x3.1mm_ThermalVias	1	₹ 548.70	₹ 548.70	**evelta.com**
9	**SMBJ60CA** D3	60V 600W Surface Mount Bidirectional TVS Diode (High-Voltage Pack Input Surge Suppression)	Diode_SMD:D_SMB	1	₹ 15.00	₹ 15.00	**evelta.com**
10	**SMAJ6.0A** D20	6.0V 400W Fast Unidirectional TVS Diode (5V Isolated DC-DC Bus Rail Clamp)	Diode_SMD:D_SMA	1	₹ 4.15	₹ 4.15	**evelta.com**
11	**BCX56-135** Q21	Nexperia 80V 1A Medium-Power NPN Bipolar Transistor (Linear Pre-Regulator Pass Element)	Package_TO_SOT_SMD:SOT-89-3_Handsoldering	1	₹ 11.38	₹ 11.38	**evelta.com**
12	**SMAJ75A** D40	75V 400W Fast Unidirectional TVS Diode (Precharge Resistor Inductive Clamp, DO-214AC)	Diode_SMD:D_SMA	1	₹ 3.30	₹ 3.30	**evelta.com**
13	**BAT54S** D42	Dual Schottky Barrier Diode Series Pair (Precharge Sense ADC Rail Clamp, SOT-23)	Package_TO_SOT_SMD:SOT-23	1	₹ 2.01	₹ 2.01	**evelta.com**

Page 1 Subtotal (LionCircuits, Robu.in, Evelta - 13 Items, 14 pcs): ₹ 2,254.93

16S LiFePO4 BMS - External Procurement (Sharvi, Rajiv Electronics & Sourcing Rules)

Power Resistors, Interconnect Sockets, Mating Cable Plugs, High-Current Terminal Hardware, and Sourcing Guidelines

#	Component & Ref	Component Description	Component Footprint	Qty	Cost Online	Total Cost	Website
Vendor: Sharvi Electronics (sharvielectronics.com) - Chassis Mount Wirewound Power Resistor							
14	**47 Ω 25W (or 50W)** R_PRE	47 Ohm (or 50 Ohm) 25W / 50W Aluminum Housed Chassis Mount Power Resistor (Precharge Inrush Limiter)	Chassis_BoltMount (Wired to J35 Precharge Header)	1	₹ 184.88	₹ 184.88	**sharvielectronics.com**
Vendor: Rajiv Electronics (rajivelectronics.com) - Industrial D-Sub Connectors, Mating Harness and Hardware							
15	**DD50_Female** J30	DD-50 50-Pin D-Sub High-Density Female PCB Connector Socket (16-Cell Sense and Balance Taps)	Connector_Dsub:DSUB-50_Female_Straight	1	₹ 112.10	₹ 112.10	**rajivelectronics.com**
16	**DB25_Female** J20	DB-25 25-Pin D-Sub Female Right-Angle PCB Connector Socket (16x Cell NTC Temperature Taps)	Connector_Dsub:DSUB-25_Socket_Horizontal_P 2.77x2.84mm_EdgePinOffset4.94mm	1	₹ 26.00	₹ 26.00	**rajivelectronics.com**
17	**DB50 Male Plug + Hood** P30	50-Pin D-Sub Male Solder Cable Plug with Metalized Shielded Hood (Mating Cable Harness for J30)	Cable_Mount_Dsub_50	1	₹ 95.00	₹ 95.00	**rajivelectronics.com**
18	**DB25 Male Plug + Hood** P20	25-Pin D-Sub Male Solder Cable Plug with Protective Metal Hood (Mating Cable Harness for J20)	Cable_Mount_Dsub_25	1	₹ 35.00	₹ 35.00	**rajivelectronics.com**
19	**FROM SHUNT (Bolt Terminal)** J32	Direct Bolt-Down Terminal Pad (M5/M6 SS Screw, Spring/Flat Washer & Flange Nut - Shunt Return)	MountingHole:MountingHole_8.4mm_M8_Pad	1	₹ 12.00	₹ 12.00	**rajivelectronics.com**
20	**PACK MINUS (Bolt Terminal)** J33	Direct Bolt-Down Terminal Pad (M5/M6 SS Screw, Spring/Flat Washer & Flange Nut - Pack Negative)	MountingHole:MountingHole_8.4mm_M8_Pad	1	₹ 12.00	₹ 12.00	**rajivelectronics.com**
21	**PACK PLUS TAP (Bolt Terminal)** J34	Direct Bolt-Down Terminal Pad (M5/M6 SS Screw, Spring/Flat Washer & Flange Nut - Pack Positive)	MountingHole:MountingHole_8.4mm_M8_Pad	1	₹ 12.00	₹ 12.00	**rajivelectronics.com**
GRAND TOTAL (All 21 External Line Items Across 5 Vendors)				22 pcs		**₹ 2,743.91***	**5 Vendors**

Vendor Sourcing and Fulfillment Quick Reference

Vendor Domain	Components Supplied	Stock Status	MOQ / Pack Rule	Vendor Subtotal
lioncircuits.com	TI BQ76952PFBR 16S Battery Monitor, Protector and Balancer IC (TQFP-48)	Verified In Stock	MOQ: 1 pc	₹ 665.29
robu.in	FL-2B 100A Shunt, FM24CL64 FRAM, UCC27517 Drivers (2x), B1209S DC-DC, PESD2CAN	Verified In Stock	MOQ: 1 pc	₹ 775.00
evelta.com	ISO1050 Isolated CAN, LM5164 Buck, SMBJ60CA, SMAJ6.0A, BCX56, SMAJ75A, BAT54S	Verified In Stock	MOQ: 1 pc	₹ 814.64
sharvielectronics.com	47 Ohm (or 50 Ohm) 25W / 50W Aluminum Housed Chassis Mount Power Resistor (Precharge)	Verified In Stock	MOQ: 1 pc	₹ 184.88
rajivelectronics.com	DD-50 & DB-25 PCB Sockets, Mating Plugs + Hoods, M5/M6 Bolt Terminals (3x)	Verified In Stock	Design: 7 pcs | Pack: 14 pk	₹ 304.10 / ₹ 448.10*

Engineering Procurement Notes & Vendor Guidelines:
1. Strict Supplier Boundary: Components listed above are exclusively procured from the user-approved Indian distributors (LionCircuits, Robu.in, Evelta, Sharvi Electronics, Rajiv Electronics). All remaining system components (MOSFETs, MCU, multiplexers, optocouplers, display, passives) are consolidated on ET Store (etstore.in).
2. Precharge Subsystem: Sheet 08 precharge path components include SMAJ75A (D40) and BAT54S (D42) on Evelta, external chassis mount power resistor R_PRE (47 ohm 25W/50W) on Sharvi Electronics, and Q19 (IRFB4110), U40 (EL817), D41 (BZT52C15), J35 (AK300-2 5.08mm) on ET Store.
3. *Order Total vs Design Qty: Rajiv Electronics M5/M6 bolt terminal hardware (J32, J33, J34) has a standard retail pack size of 5 units. Ordering full packs yields ₹448.10 at Rajiv Electronics, bringing the total order cost across all 5 non-ETStore vendors to ₹2,887.91.
4. Verification & Traceability: All parts are verified for active stock status with pin-compatible footprints in the 16S BMS KiCad EDA project.